

Agents and Routing Quick Reference

The vocabulary from Session 6, on one card

The one definition to memorise

Agent equals LLM plus tools plus knowledge plus memory plus autonomous action. Drop the autonomous action and it is not an agent, however many tools it calls. Every term here also appears in the Session Notes with the example it came from.

1. Concepts

Term	What it means
Action	The high level goal you hand the agent. "Build me a chair."
Tool calling	One step the LLM takes toward that goal. The hammer and the saw. One action can need many tool calls.
Agency (autonomy)	How freely the agent may act. Moves inversely to predictability.
Guardrail	The boundary that keeps autonomy safe. The rope on the cow. Not optional.
Workflow	A non agentic application: LLM calls through predefined code paths, each with one task. A news summariser is a workflow.
ReAct	Reason, act, observe, repeat. Ends when the agent stops calling tools.

2. Building the Agent

Term	What it means
Tool decorator	Registers a Python function in LangChain's tool registry. Registration never executes it.
Docstring	How the model learns what a tool does. Written for LLMs now. A wrong one misleads the agent.

Term	What it means
bind_tools	Writes tool names, descriptions and shapes into the system prompt. The model can only pick from what you bind.
CRAFT prompt	Context, role, action, format, task. Give it a role, set ground rules, state the task, specify the output.

3. Memory and Routing

Term	What it means
Short term memory	One session, like a single ChatGPT window. Uses checkpoints. Like RAM.
Long term memory	Persists across sessions. Uses memory stores. Like a hard disk.
Checkpoint	Saves and restores agent state. In memory saver for experiments, SQLite or Postgres for persistence.
Thread ID	The session identifier passed in config. Same idea as the UUID in a ChatGPT URL. Your backend generates it.
Semantic routing	Picking a workflow or agent from the meaning of the query, using embeddings not an LLM. Also called intent based routing.
Route	A name plus example queries. One per business intent or workflow, never one per tool.
Utterance	An example query attached to a route. What gets embedded and stored. Router quality lives here.
Threshold	Cosine similarity a query must reach to match. Defaults to 0.5, tuned per route. Lower accepts more and misroutes more.
Default route	Where a query goes when it matches nothing. Stops off topic questions breaking things.

Two checks worth running

Ask honestly whether what you build is an agent or a workflow: if nothing autonomous happens, call it a workflow. And when routing sits in front of your RAG pipelines, the integration is a plain if else on the route name. The clever part is the utterances, not the plumbing.